Team Project Abstract

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SWPP Practice Session

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Team Project

- Develop compiler optimizations as a team!
- Each team should work on a team repository
  - Each team should create an upstream repository swpp202601-teamN
  - Each member should work on individual fork of the upstream
  - … Unless you are the owner of the repository

Collaboration Basics

- Discuss what to implement with teammates
- Implement new feature/bugfix/etc on a new branch
- Send a pull request (PR) to the upstream repository
- Get the PR reviewed by at least 2 reviewers
- Merge the PR into the upstream iff it passes the review

Discussion

- Share your plans before getting into work!
  - Helps teammates understand your work
  - Prevents duplicate works between teammates
- Use GitHub Issue to keep a record of each member's plans
- Presenting short slides can be helpful
  - Especially if you're planning on a very complex algorithm…

Pull Request

- A unit of working feature
- After each PR is merged…
  - The project must compile without any problem
  - All existing tests must pass
- PRs are ‘destined’ to be reviewed
  - Measures should be taken to reduce the burden of reviewing!

Line Diff

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- ~~Generally speaking, PRs should not introduce 200+ line diff~~
  - ~~This does not include comments or tests~~
- But we have the power of AI in our hand!
  - We should be able to manage bigger diffs...

Reducing the Diff

- Line diffs should be reduced as much as possible
- One can reduce the line diff by…
  - Splitting a large PR into multiple PRs
  - Writing concise and expressive codes
  - Following the formatting rules

Splitting the Pull Request

- Each PR is responsible for a single subject
- Common mistakes that ‘bloat’ the PRs are…
  - Implement A and format B
  - Fix irrelevant bugs A and B
  - Implement a large feature A and fix B to use A

Splitting the Pull Request

- Sometimes, a single subject may seem to require large diff
- A feature that requires complex algorithm for efficiency
  - Start with simpler implementation, then improve it with another PR
- Fixing tightly related bugs A, A′ and A″
  - Fix A first, then fix A′ and A″ based on merged changes

Concise & Expressive Codes

- This topic is somewhat language-specific (C++)
- Use `auto`, `decltype`, or `using statement` to hide complex types
- Look for library functions (`std::do_this()`, `llvm::doThat()`)
- Follow commonly used design patterns (visitor, builder, ...)
  - Template-based design patterns may be hard to use
  - But they are very powerful! (CRTP, variadic templates, type traits, ...)

Code Formatting

- Properly formatted code is easier to read and understand
- Also, formatting reduces diffs due to minor editing
  - Spaces, newlines, parentheses, etc
- `clang-format` can be used to apply formatting rules file-wide
  - Already installed in LLVM bin directory

Code Review

- Reviewers (teammates) will take time understanding your code
- This may look slower compared to writing the code alone
- But this process makes the code less buggy & simpler
- In the long run, you get a better program in shorter time ☺

Code Review

- Based on their understanding, reviewers may
  - Approve your code to be merged
  - Give feedbacks to suggest changes
  - Ask questions about the implementation

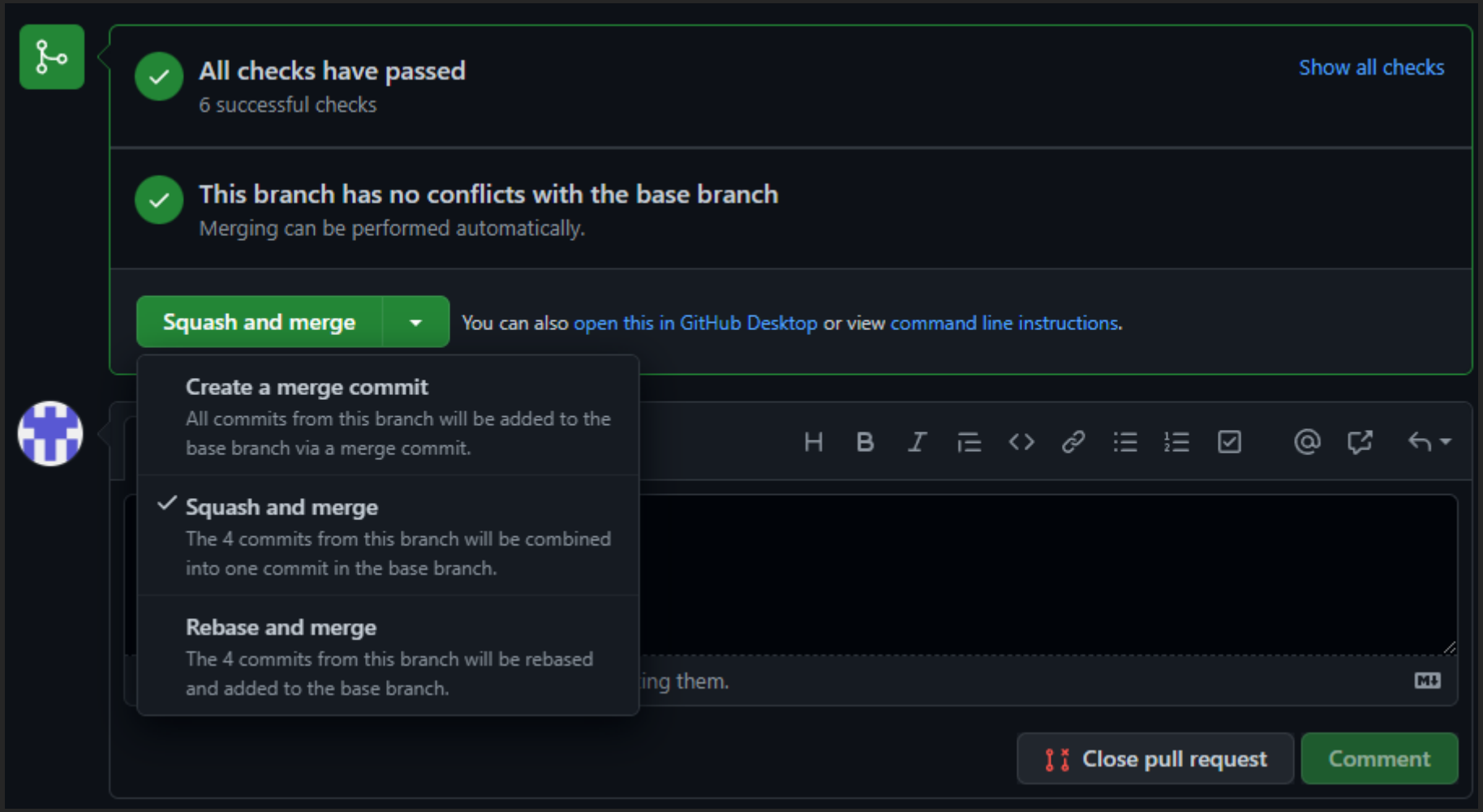
Giving Feedbacks

- Reviewers should look for the following criteria
  - Does this PR correctly address the issue?
  - Does this PR not introduce any UB or unnecessary copy?
  - Does this PR include proper tests to show the correctness?
  - Is there any existing library that does the same job?
  - Is there any better idiom or pattern that has the same meaning?

Code Review

- Feedbacks may be heartbreaking ☹
  - Remember that reviewers are not blaming you for bad code
  - Code review is an act of collaboration toward better code
- Answer the questions for you and your teammates
  - Better understanding leads to better feedbacks
  - You may ‘rubber-duck’ unnoticed problems in the process

Merging the Pull Request

- Use squash and merge
  - Makes the commit history concise & linear
  - Keeps the ‘intermediate commits’ from flooding the upstream
  - Helps reduce buggy PR into buggy commit

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